Solutions: Problem Set 3

Version of October 20, 2026

Problem 3.1 (Planning a stratified sample). We *plan* a survey to estimate the population mean of environmental expenditures (`expend`). We want to assess whether stratification by industry group (`code`) can improve precision relative to a simple random sample without replacement (SRS).

We use the data set `sepe.dat` from Problem Set 1. Treat the data as an SRS of $n = 800$ establishments from a population of size $N = 3770$. We focus on industry class 31, that is, the domain U_{31} of size $N_{31} = 650$, identified with `code == 1`. Assume that:

- the planned net sample size is $n = 800$ establishments;
- the anticipated response rate is 80 % in both strata; and
- the within-stratum variances of `empl` and `expend` can be estimated from the existing SRS `sepe.dat`.

a) Allocate the net sample *proportionally* to the two strata. Determine the required gross sample sizes and estimate the variance of the stratified estimator of the population mean of expenditures.

Solution. The population sizes are $N_{rest} = 3120$ for `code == 0` and $N_{31} = 650$ for `code = =1`, so $N = 3770$.

```
> sepe <- read.table("https://stat.ethz.ch/Teaching/Datasets/cas-das/sepe.dat",
                     header = TRUE)

> Nh <- c("rest" = 3120, "31" = 650)
> N <- sum(Nh)
> Wh <- Nh / N
> Wh

     rest        31
0.8275862 0.1724138
```

Under proportional allocation,

$$n_h \approx n\, W_h, \qquad W_h = \frac{N_h}{N}.$$

We round the stratum sample sizes while keeping their sum equal to $n = 800$.

```
> n <- 800
> n_prop <- round(n * Wh)
> n_prop["rest"] <- n - n_prop["31"]
> n_prop

rest   31
 662  138
```

Because the anticipated response rate is the same in both strata, the gross sample can also be allocated proportionally. The total gross sample size is $800/0.8 = 1000$.

```
> response_rate <- 0.8
> n_gross <- round(n / response_rate)
> n_prop_gross <- round(n_gross * Wh)
> n_prop_gross["rest"] <- n_gross - n_prop_gross["31"]
> n_prop_gross
```

```
rest   31
 828  172
```

Thus, the required gross sample sizes are 828 establishments with `code == 0` and 172 with `code == 1`.

The variance of the stratified mean $\bar{y}^{\text{strat}} = \sum_{h=1}^{H} W_h \, \bar{y}_{h,U}$ is

$$\text{var}_p(\bar{y}^{\text{strat}}) = \sum_{h=1}^{H} W_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{\sigma_h^2}{n_h}.$$

We estimate σ_h^2 using the existing simple random sample.

```
> sigma2_y_h <- tapply(sepe$expend, sepe$code, var)
> var_prop <- sum(Wh^2 * (1 - n_prop / Nh) * sigma2_y_h / n_prop)
> var_prop
```

```
[1] 4423859
```

The estimated variance under proportional allocation is therefore 4'423'859.

> **Remarks.** The response rate affects the *gross* sample size, but the variance calculation is based on the planned net sample sizes. Using the general variance formula also avoids any approximation caused by rounding the proportional allocation.

b) Now use a Neyman-type (optimum) allocation, but base the allocation on the within-stratum standard deviations of the auxiliary variable `empl` rather than on `expend`.

- Determine the required gross sample sizes.
- Under what conditions is it reasonable to derive the allocation from an auxiliary variable?
- Estimate the variance of the stratified estimator of mean environmental expenditures under this allocation.

Solution. The Neyman allocation for a target variable y is

$$n_h = n \, \frac{W_h \, \sigma_{y,h}^2}{\sum_{j=1}^{H} W_j \, \sigma_{y,j}^2}, \qquad h = 1, \ldots, H.$$

Here we deliberately replace $\sigma_{y,j}^2$ by the (estimates of) within-stratum standard deviations of `empl`.

```
> sigma_x_h <- tapply(sepe$empl, sepe$code, sd)
> names(sigma_x_h) <- c("rest", "31")
> alloc_x <- Wh * sigma_x_h
```

```
> n_aux <- round(n * alloc_x / sum(alloc_x))
> n_aux

rest   31
 629  171

> n_aux["rest"] <- n - n_aux["31"]
> n_aux

rest   31
 629  171
```

The corresponding gross allocation is

```
> n_aux_gross <- round(n_gross * alloc_x / sum(alloc_x))
> n_aux_gross["rest"] <- n_gross - n_aux_gross["31"]
> n_aux_gross

rest   31
 787  213
```

An auxiliary-variable allocation is useful when the pattern of within-stratum variability of the auxiliary variable resembles that of the target variable. More precisely, we would like $\sigma_{y,h} \approx c \cdot \sigma_{x,h}$ with approximately the same positive constant c across strata. In that case, the allocation based on x is close to the Neyman allocation for y. Historical values of the target variable are a common example of a useful auxiliary variable.

The existing sample can be used to inspect this condition directly:

```
> sigma_y_h <- sqrt(sigma2_y_h)
> cbind(sigma_x_h = sigma_x_h, sigma_y_h = sigma_y_h,
        ratio = sigma_y_h / sigma_x_h)

     sigma_x_h sigma_y_h    ratio
rest   177.936  62525.73 351.3945
31     231.377  85396.08 369.0776
```

The allocation was computed from `empl`, so its variance must be evaluated using the *actual* stratum sample sizes above and the within-stratum variances of `expend`:

```
> var_aux <- sum(Wh^2 * (1 - n_aux / Nh) * sigma2_y_h / n_aux)
> var_aux

[1] 4332910
```

The estimated variance is therefore 4'332'910.

c) Compare the estimated variances from parts a) and b) with the estimated variance under a simple random sample of $n = 800$. Does stratification improve precision?

Solution. Let $\bar{y}_{\text{SRS}}$ denote the sample mean for SRS. It holds that

$$\widehat{\text{var}}_p(\bar{y}_{\text{SRS}}) = \left(1 - \frac{n}{N}\right) \frac{\sigma_y^2}{n},$$

where σ_y^2 is estimated from `sepe.dat`.

```
> var_srs <- (1 - n / N) * var(sepe$expend) / n
> var_srs
```

```
[1] 4567134
```

The three planning variances are:

```
> c(SRS = var_srs, proportional = var_prop, auxiliary = var_aux)
```

```
         SRS proportional    auxiliary
     4567134      4423859      4332910
```

Relative to simple random sampling, the estimated variance reductions are

```
> 100 * c(proportional = 1 - var_prop / var_srs,
          auxiliary = 1 - var_aux / var_srs)
```

```
proportional    auxiliary
    3.137084     5.128461
```

percent. The auxiliary-variable allocation improves on proportional allocation if $\widehat{\mathrm{var}}_{p_{\mathrm{aux}}} < \widehat{\mathrm{var}}_{p_{\mathrm{prop}}}$. Unlike a true Neyman allocation based on `expend`, this ordering is not guaranteed because the allocation in part b) is based on `empl`.

Problem 3.2 (Weighing Basu's elephants). A circus owns 50 elephants.[1] The circus owner wants to estimate their total weight without having to weigh every elephant. He believes that Timbo is approximately as heavy as an average elephant and proposes weighing Timbo and multiplying his weight by 50, formally

$$\widehat{t}_y^{\,\mathrm{owner}} = 50 \cdot y_{\,\mathrm{Timbo}}.$$

A statistician insists on probability sampling. They agree to select one elephant at random, with the following sampling design:

$$p(s) = \begin{cases} 0.99, & \text{if Timbo is selected,} \\ 1/4900, & \text{otherwise.} \end{cases}$$

The statistician proposes using the Horvitz–Thompson estimator, $\widehat{t}_y^{\,\mathrm{HT}}$, of the population total.

a) Express the Horvitz–Thompson estimator in terms of the selected elephant's weight and its selection probability. Show that the estimator is design-unbiased.

[1] This problem is adapted from Basu's famous elephant example in survey sampling; see Basu, D. (1971). An Essay on the Logical Foundations of Survey Sampling, Part 1, in: *Foundations of Statistical Inference*. Proceedings of the Symposium on the Foundations of Statistical Inference Prepared Under the Auspices of the René Descartes Foundation and Held at the Department of Statistics, University of Waterloo, Ontario, Canada, from March 31 to April 9, 1970 ed. by Godambe, V. P. and Sprott, D. A. Toronto/ Montreal: Holt, Rinehart and Winston of Canada, pp. 203–233. The population data are taken from the book: Brewer, K. (2002), *Combined Survey Sampling Inference: Weighing Basu's Elephants*, London: Arnold, Appendix B.

Solution. Let I denote the index of the selected elephant. The HT estimator is

$$\widehat{t}_y^{\mathrm{HT}} = \begin{cases} \dfrac{y_{\text{Timbo}}}{0.99}, & \text{if Timbo is selected,} \\ 4900 \cdot y_I, & \text{otherwise.} \end{cases}$$

Its design expectation is

$$\mathbb{E}_p\big(\widehat{t}_y^{\mathrm{HT}}\big) = \sum_{i \in U} p(\{i\}) \frac{y_i}{p(\{i\})} = \sum_{i \in U} y_i = t_y.$$

Hence, the HT estimator is design-unbiased.

b) Suppose Timbo is selected and weighs 3 900 kg. Calculate the estimates obtained by the circus owner and the statistician. Discuss the two estimates. Does design-unbiasedness guarantee that an estimator will give a reasonable estimate for the observed sample?

Solution. Set

```
> y_timbo <- 3900
```

The owner's estimator gives

```
> t_owner <- 50 * y_timbo
> t_owner
```

```
[1] 195000
```

whereas the HT estimator yields

```
> t_HT <- y_timbo / 0.99
> t_HT
```

```
[1] 3939.394
```

The estimates differ substantially. The owner's estimate is 195 000 kg, whereas the HT estimate is only about 3 939 kg, roughly the weight of a single elephant. Design-unbiasedness does not guarantee an accurate estimate for a particular sample.

c) Ken Brewer provides the actual weights of all 50 elephants in the file:

```
> elephants <-
    read.table("https://stat.ethz.ch/Teaching/Datasets/cas-das/elephants.dat",
               header = TRUE)
```

We can use these data to investigate the sampling distribution of the HT estimator. Construct the sampling distribution of the HT estimator by calculating its possible values and their probabilities. Also, calculate the design expectation and standard deviation of the HT estimator. Discuss the results.

Solution. There are 50 possible samples, each consisting of one elephant. We calculate the sampling probabilities and the corresponding HT estimates.

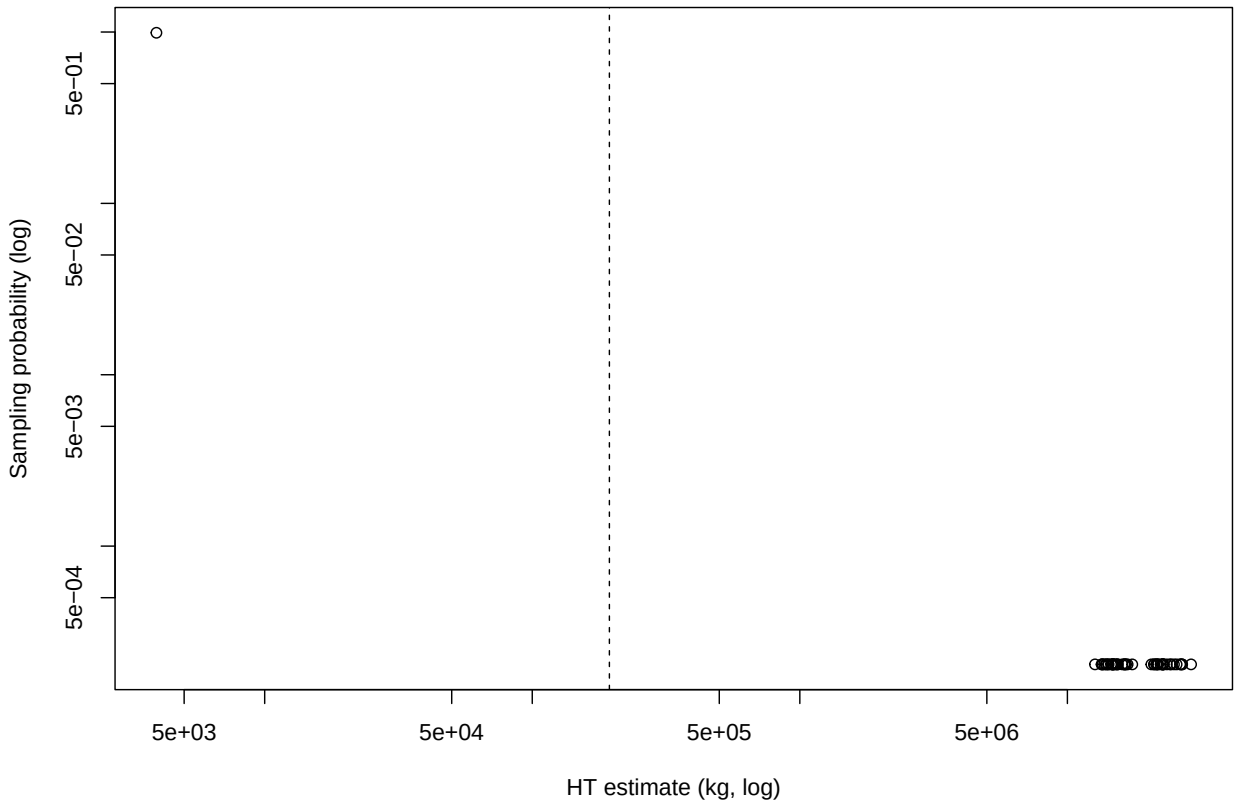


Figure 1: Samplign distribution (axes in logarithmic scales; the dashed line shows the true population total)

```
> p <- ifelse(elephants$name == "Timbo", 0.99, 1/4900)
> t_HT <- elephants$weight / p
> sampling_dist <- data.frame(name = elephants$name, estimate = t_HT,
                              probability = p)
```

The complete sampling distribution can be visualized by plotting the possible estimates against their probabilities; see Figure 1. We use logarithmic scales because the estimates and probabilities differ substantially in magnitude. With probability 0.99, the estimator yields only about 3,939 kg. The remaining probability of 0.01 is distributed over extremely large estimates, ranging from approximately 12.7 to 29.0 million kg. These rare outcomes compensate for the severe underestimation when Timbo is selected.

```
> plot(t_HT, p, log = "xy", xlab = "HT estimate (kg, log)",
       ylab = "Sampling probability (log)")
> abline(v = sum(elephants$weight), lty = 2)
```

The HT estimator is design-unbiased:

```
> E_HT <- sum(p * t_HT)
> identical(E_HT, as.numeric(sum(elephants$weight)))
```

```
[1] TRUE
```

The design standard deviation is approximately 1.95 million kg, about ten times the population total.

```
> SD_HT <- sqrt(sum(p * (t_HT - E_HT)^2))
> SD_HT
[1] 1950266
```

Problem 3.3 (PPS Sampling and the Horvitz–Thompson estimator). We use the data set `sepe.dat` from Problem Set 1 and treat the $N = 800$ establishments as the finite population U. We want to draw a PPS sample without replacement of size $n = 200$, using the number of full-time employees as the measure of size. The data set contains:

`empl`: number of full-time employees, denoted by x_i. These values are known from a register before sampling;

`expend`: environmental expenditures, denoted by y_i. We treat these values as unknown until an establishment is sampled.

a) Compute the first-order inclusion probabilities and draw a PPS sample of $n = 200$ establishments without replacement using Brewer's method. If there were no certainty units, the inclusion probabilities would be

$$\pi_i = \frac{n x_i}{t_x}, \qquad t_x = \sum_{i \in U} x_i.$$

Compare these naive probabilities with the probabilities returned by `pps_probabilities()` of the `robsurvey` package. What do you observe?

Hint. Attach the packages `survey` and `robsurvey` to the search path. Compute the inclusion probabilities and draw the sample as follows:

```
pps_setup <- pps_probabilities(size = sepe$empl, n = 200)
pi <- as.numeric(pps_setup)

set.seed(11)
s <- pps_draw(pps_setup, method = "brewer")
```

Solution. We attach the packages to the search path and load the data:

```
> library("survey")
> library("robsurvey")
> sepe <- read.table("http://stat.ethz.ch/Teaching/Datasets/WBL/sepe.dat",
                     header = TRUE)
```

The valid first-order inclusion probabilities are

```
> pps_setup <- pps_probabilities(sepe$empl, n = 200)
> sepe$pi <- as.numeric(pps_setup)
> summary(sepe$pi)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.07253 0.18082 0.25000 0.36296 1.00000
> sum(sepe$pi == 1)
```

```
[1] 16
```

Hence, 16 establishments are certainty units with $\pi_i = 1$. By contrast, the naive formula gives

```
> pi_naive <- 200 * sepe$empl / sum(sepe$empl)
> summary(pi_naive)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.07102 0.17706 0.25000 0.35543 1.55791
```

```
> sum(pi_naive > 1)
```

```
[1] 15
```

Some of these values exceed one and therefore cannot be inclusion probabilities. The function `pps_probabilities()` handles this iteratively: certainty units are assigned $\pi_i = 1$, and the probabilities of the remaining units are rescaled so that

$$0 < \pi_i \leq 1 \qquad \text{and} \qquad \sum_{i \in U} \pi_i = n.$$

Thus, with certainty units present, the final probabilities are not simply nx_i/t_x for every unit. We then draw the PPS sample:

```
> set.seed(11)
> s <- pps_draw(pps_setup, method = "brewer")
> length(s)
```

```
[1] 200
```

b) Create a survey design object for the sample from part a). Then use `svyplot()` to plot environmental expenditures against the number of employees. The bubble sizes represent the sampling weights $w_i = 1/\pi_i$. Describe the main features of the plot.

Remarks. For the PPS design of part a), use

```
dn_sepe <- svydesign(ids = ~1, fpc = ~pi, pps = "brewer",
                     data = sepe[s, ])
```

Here, `fpc = ~pi` supplies the first-order inclusion probabilities. The option `pps = "brewer"` tells `survey` to use Brewer's approximation for the second-order inclusion probabilities needed for variance estimation. This is distinct from Brewer's sampling algorithm used in part a).

Solution. We define the design by

```
> dn_sepe <- svydesign(ids = ~1, fpc = ~pi, pps = "brewer",
                       data = sepe[s, ])
```

Omitting `fpc` would instead create a with-replacement approximation to the design.

The scatterplot (Figure 2) is obtained with the commands:

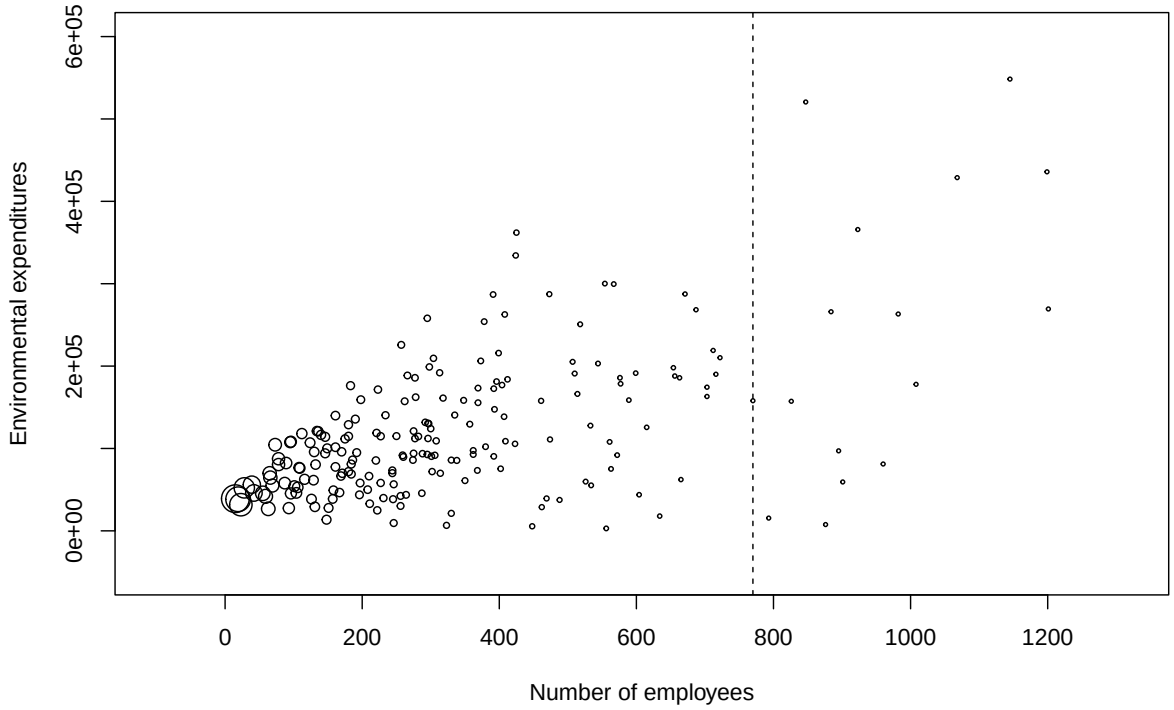


Figure 2: Environmental expenditures vs. number of employees; the size of the bubbles is proportional to the sampling weights

```
> svyplot(expend ~ empl, dn_sepe, inches = 0.1,
          xlab = "Number of employees",
          ylab = "Environmental expenditures")
> certainty_cut <- min(sepe$empl[sepe$pi == 1])
> abline(v = certainty_cut, lty = 2)
```

The main features are:

- Establishments with fewer employees have smaller inclusion probabilities and therefore larger sampling weights. They appear as larger bubbles.
- The largest establishments are certainty units. The dashed line marks the smallest value of `empl` among these units.
- The size measure `empl` is positively related to `expend`: 0.57. PPS sampling is particularly effective when y_i is closely related to the size measure, so that the ratios y_i/x_i do not vary too much.

c) Estimate the population total of environmental expenditures,

$$t_y = \sum_{i \in U} y_i,$$

using the Horvitz–Thompson estimator and the design from part b). Also compute an estimate of the standard error of the estimator.

Solution. The Horvitz–Thompson estimate and its estimated standard error are obtained directly with

```
> t_y <- svytotal(~expend, dn_sepe)
> t_y
```

```
          total      SE
expend 71293708 3300926
```

The PPS weights have an additional useful property. Because the inclusion probabilities were constructed from `empl`, the Horvitz–Thompson estimator reproduces its known population total:

```
> t_x <- svytotal(~empl, dn_sepe)
> c(estimate = coef(t_x), population = sum(sepe$empl))
```

```
estimate.empl    population
       154181        154181
```

The point estimate is therefore exact, but the default variance estimator does not automatically use this calibration information. We can tell `survey` that the supplied weights are already calibrated to t_x:

```
> dn_sepe_calib <- svydesign(ids = ~1, fpc = ~pi, pps = "brewer",
                             data = sepe[s, ],
                             calibrate.formula = ~ -1 + empl)
> svytotal(~empl, dn_sepe_calib)
```

```
      total SE
empl 154181  0
```

The formula has no intercept because the PPS weights reproduce t_x, but they do not generally reproduce the population size N. The option `calibrate.formula` does not change the weights; it tells the variance estimator about an existing calibration equation.

The estimate of t_y is unchanged, while its estimated standard error can decrease:

```
> t_y_calib <- svytotal(~expend, dn_sepe_calib)
> c(estimate = unname(coef(t_y_calib)), SE_without_calibration = SE(t_y),
    SE_with_calibration = SE(t_y_calib))
```

```
            estimate SE_without_calibration    SE_with_calibration
            71293708               3300926                3313698
```